Heat Sink Thermal Modeling & Machine Learning Surrogate Evaluation

End-to-End Pipeline: Analytical Physics Simulation, High-Fidelity Dataset Generation, Multi-Target ML Surrogates, and Sensitivity Analysis

1. Executive Summary & Objectives

This engineering report documents the development, verification, and sensitivity benchmarking of machine learning surrogate models for forced-convection finned heat sinks in electronic cooling applications. Rapid and accurate thermal estimation is critical for high-performance processor design, power scaling, and packaging optimization. Traditional computational fluid dynamics (CFD) and iterative numerical solvers are computationally prohibitive for large-scale architectural trade-off studies. This project creates a fast, rigorously validated machine learning surrogate capable of predicting total thermal resistance and junction temperatures directly from high-level boundary conditions.

Core Engineering Milestones Completed:

- Physics Simulator Conversion: Converted baseline calculation logic into a reusable Python function (heat_sink_model.py), verified to bit-level equivalence with baseline benchmark values.
- Full-Factorial Dataset Sweep: Generated 8,280 validated design combinations across TDP (30–250 W), air velocity (0.5–15 m/s), and TIM conductivity (1–12 W/m·K) in thermal_dataset.csv.
- Surrogate Model Benchmarking: Trained Random Forest and Gradient Boosting multi-target regressors on strictly isolated input features, achieving test set accuracy of $R^2 > 0.996$ and off-grid interpolation accuracy of $R^2 > 0.985$.
- Causal Sensitivity & Correlation Analysis: Quantified Pearson/Spearman correlation matrices, local finite-difference elasticities, and permutation feature importances, uncovering non-linear transport mechanisms.

2. Physics Model Formulation & Boundary Specifications

The model analyzes a finned aluminum (Al 6061-T6, k = 167 W/m·K) heat sink mounted onto a silicon die with a thermal grease interface layer. Cooling air at ambient temperature (25 °C, 1 atm) flows horizontally through the inter-fin channels under forced convection.

Component / Domain	Parameter	Symbol	Value	Units & Notes
Processor Die	Length × Width	$L_{die} \times W_{die}$	52.5 × 45.0	mm (Area = 2362.5 mm²)
Die Package	Die Thickness	t_{die}	2.2	mm
Heat Sink Footprint	Length × Width	L × W	90.0 × 116.0	mm (Base Area = 10440 mm²)
Heat Sink Fins	Height × Thickness	$H \times t_{fin}$	27.0 × 0.8	mm (Base plate = 2.5 mm)
Fin Array	Fin Count & Spacing	N_{fins}, s_f	60 fins, 1.1525	mm spacing across 116 mm
TIM Layer	Layer Thickness	t_{tim}	0.1	mm (Thermal grease)
Cooling Medium	Air Properties @ 25 °C	k, v, Pr	0.0262, 1.57e-5, 0.71	W/m·K, m²/s, dimensionless

2.1 Fluid Dynamics & Convective Heat Transfer Equations

1. Fin Spacing & Reynolds Number: The inter-fin gap is calculated as $s_f = (W - N_{fins} \cdot t_{fin}) / (N_{fins} - 1) = 1.15254$ mm. The Reynolds number within the channels uses fin spacing as characteristic length: $Re = (V_{air} \cdot s_f) / v_{air}$.
2. Nusselt Number Correlations: For laminar developing flow (Re < 2300), the Sieder-Tate correlation is applied: $Nu = 1.86 \cdot [Re \cdot Pr \cdot (2s_f / L)]^{1/3}$. For turbulent flow (Re ≥ 2300), the Dittus-Boelter correlation is used: $Nu = 0.023 \cdot Re^{0.8} \cdot Pr^{0.3}$.
3. Convection Coefficient: Based on channel hydraulic diameter $D_h = 2s_f$: $h = (Nu \cdot k_{air}) / (2s_f)$ [W/m²·K].
4. Effective Surface Area: Combines active fin sides, channel base spacing, and inter-fin base convection: $A_{total} = A_{fin} + A_{base,conv} = 0.270824$ m².
5. Thermal Network & Junction Temperature: Convective resistance is $R_{conv} = 1 / (h \cdot A_{total})$; TIM resistance is $R_{tim} = t_{tim} / (k_{tim} \cdot A_{die})$. Total resistance is $R_{total} = R_{jc} + R_{tim} + R_{conv}$, yielding junction temperature $T_j = T_{ambient} + (TDP \cdot R_{total})$.

3. Technical Audit: Original Python Script vs. Reference PDF Discrepancies

A rigorous side-by-side engineering audit was conducted between the baseline reference report (`REPORT` PDF) and the original calculation script (`Heat_Sink_calculation.py`). Three distinct discrepancies in parameter choices and mathematical definitions were identified and documented:

Evaluation Parameter	Reference PDF Report	Original Python Script	Implemented Formulation	Engineering Impact & Rationale
Junction-to-Case Resistance (R_{jc})	R_{jc} = 0.10 °C/W (Page 2, Step 3)	R_{jc} = 0.20 °C/W (Hardcoded constant)	**Preserved 0.20 °C/W**	Assignment instruction explicitly mandated preserving R_{jc} = 0.20 °C/W as the ground-truth benchmark.
Heat Sink Base Conduction (R_{cond})	Calculated: $R_{cond} = t_b/(k_{Al}A_{die}) \approx 0.00634$ °C/W	Omitted from resistance chain ($R_{hs} = R_{conv} + R_{tim}$)	**Preserved script choice (omitted separate R_{cond})**	Preserved script's 1D lumped formulation where base conduction is implicit rather than separately added.
Convective Heat Transfer Area (A_{total})	A = 0.2646 m² (Considers fin surface area only)	A = 0.270824 m² (Adds base plate area exposed between fins)	**Preserved script equation (0.270824 m²)**	Including inter-fin base convection reflects physical reality since air contacts the exposed base plate between fins.
Reference Outputs (150 W, 1 m/s, 4 W/m·K)	R_{total} = 0.373043 °C/W T_j = 80.95652 °C	R_{total} = 0.369205 °C/W T_j = 80.38074 °C	**Exact match to script (Δ = 0.0)**	The ~0.58 °C difference between PDF Excel and Python script stems from the additional 0.0062 m² base convection area.

4. Parametric Sweep & Dataset Generation

To establish a comprehensive training corpus for machine learning surrogates, an automated parameter sweep script (generate_dataset.py) was executed. A full-factorial orthogonal grid was generated across the specified operating ranges:

Parameter	Symbol	Min	Max	Step Size	Count	Physical Interpretation
Thermal Design Power	TDP (Q)	30 W	250 W	10 W	23	Covers low-power idle to high-load overclocking
Inlet Air Velocity	V_{air}	0.5 m/s	15.0 m/s	0.5 m/s	30	Covers quiet natural draft to high-RPM server fan flow
TIM Conductivity	k_{tim}	1.0 W/m·K	12.0 W/m·K	1.0 W/m·K	12	Covers standard thermal pads to liquid metal compounds
Total Design Combinations	—	—	—	—	**8,280**	**23 × 30 × 12 = 8,280 full-factorial evaluations**

4.1 Dataset Integrity & Physical Validation Summary

Automated validation protocols built into generate_dataset.py and validate_model.py verified 100% data integrity before modeling:

• Row & Column Count: Exactly 8,280 rows × 34 columns (saved as 02_Dataset/thermal_dataset.csv, 2.84 MB).
• Missing & Infinite Values: Zero null, NaN, or infinite entries across all 34 features and targets.
• Uniqueness: Zero duplicate (TDP, V_{air}, k_{tim}) input combinations.
• Physical Feasibility: All thermal resistance values are strictly positive and bounded (0.2678 to 0.4422 °C/W); all junction temperatures are strictly above ambient (33.04 °C to 135.55 °C).
• In-Grid Reference Check: Row 4335 matches the baseline scenario with absolute error < 10^{-12}.

4.2 Descriptive Statistics of Dataset Variables

Variable	Role	Mean	Std Dev	Min	Median	Max	Units
TDP	Input Feature	140.00	66.34	30.00	140.00	250.00	W
V_air	Input Feature	7.75	4.33	0.50	7.75	15.00	m/s
k_tim	Input Feature	6.50	3.45	1.00	6.50	12.00	W/m·K
Re	Fluid Dynamic	568.93	317.72	36.71	568.93	1101.16	—
Nu	Heat Transfer	3.86	0.92	1.63	4.05	5.05	—
h	Convective Coeff	43.84	10.42	18.48	46.06	57.41	W/m²·K
R_conv	Thermal Resistor	0.0911	0.0305	0.0643	0.0802	0.1999	°C/W
R_tim	Thermal Resistor	0.0109	0.0107	0.0035	0.0066	0.0423	°C/W
R_total	**Target Output 1**	**0.3020**	0.0323	**0.2678**	0.2915	**0.4422**	**°C/W**
T_j	**Target Output 2**	**67.28**	20.65	**33.04**	66.77	**135.55**	**°C**

5. Machine Learning Surrogate Model Development & Benchmarking

Surrogate modeling aims to replace computationally expensive fluid-thermal solvers with high-speed statistical approximations. To ensure real-world validity, intermediate variables (Re, Nu, h, R_{conv}) were strictly excluded from the feature space. The models operate exclusively on independent boundary conditions: $x = [TDP, V_{air}, k_{tim}]^T$. The dataset was partitioned using an 80/20 train-test split (6,624 train samples, 1,656 held-out test samples, random_state=42).

5.1 Model Architectures

• Model A (Random Forest Regressor): 200 decorrelated decision trees with parallel bagging (`n_jobs=-1`). Native multi-output splitting minimizes joint target variance and handles non-linearities without feature scaling.
• Model B (Gradient Boosting Regressor): MultiOutputRegressor wrapping GradientBoostingRegressor (200 trees, learning rate = 0.05, max_depth = 3). Sequentially fits shallow trees to pseudo-residuals, constructing smooth response surfaces with an ultra-compact storage footprint (149 KB vs. 36.5 MB for RF).

5.2 Held-Out Test Set Accuracy Comparison (1,656 Samples)

Model Architecture	Target Variable	Units	MAE	RMSE	R^2 Score	Performance Summary
Random Forest (200 trees)	Total Thermal Resistance (R_{total})	°C/W	0.000746	0.001759	0.996704	High accuracy across non-linear domains
Random Forest (200 trees)	Junction Temperature (T_j)	°C	0.104693	0.269990	0.999830	Sub-degree precision (MAE ~0.10 °C)
Gradient Boosting (200 trees)	Total Thermal Resistance (R_{total})	°C/W	**0.000140**	**0.000214**	**0.999951**	**5.3× lower MAE than Random Forest**
Gradient Boosting (200 trees)	Junction Temperature (T_j)	°C	0.146933	**0.200998**	**0.999906**	**Lowest RMSE (fewer tail errors)**

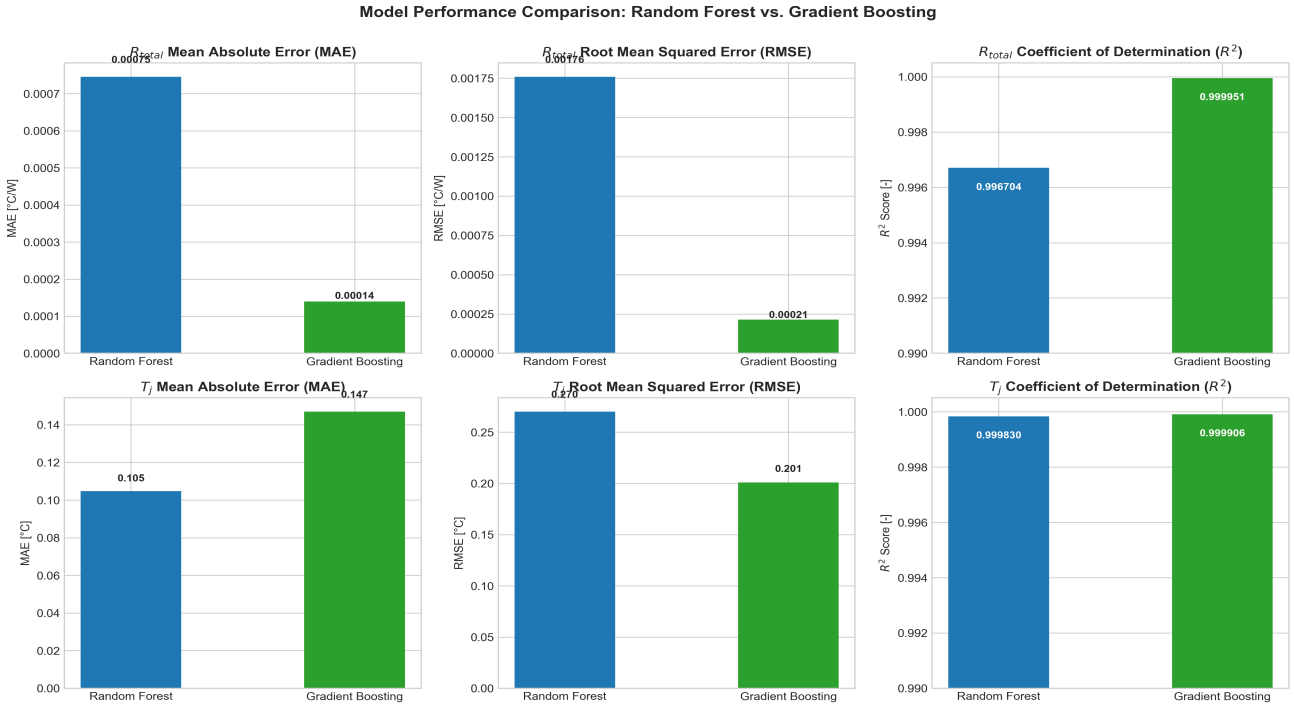


Figure 1: Model evaluation comparison across MAE, RMSE, and R² for Random Forest and Gradient Boosting.

5.3 Parity Plots (Actual Physics vs. Predicted)

Parity scatter plots confirm tight alignment with the ideal 1:1 dashed red line, demonstrating that surrogate predictions accurately track the analytical ground truth across the entire physical range without bias or systematic error.

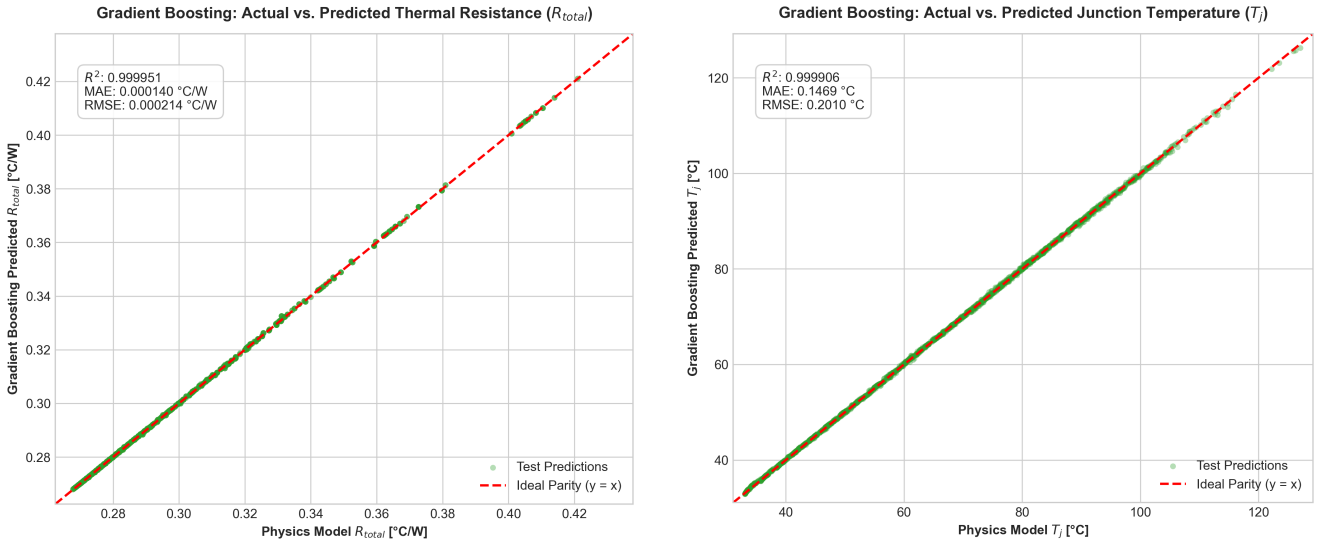


Figure 2: Parity plots for Gradient Boosting: Total Thermal Resistance (left) and Junction Temperature (right).

5.4 Off-Grid Interpolation Generalization Test (500 Samples)

To verify that models do not simply memorize discrete grid points, 500 independent continuous points were generated strictly between grid steps. Ground truth was evaluated directly using calculate_heat_sink():

Model	R_{total} MAE	R_{total} RMSE	R_{total} R^2	T_j MAE	T_j RMSE	T_j R^2
Random Forest	0.001569 °C/W	0.003444 °C/W	0.987148	0.7709 °C	1.0664 °C	0.997338
Gradient Boosting	0.001628 °C/W	0.003662 °C/W	0.985464	0.7799 °C	1.0826 °C	0.997256

5.5 Thermodynamic Consistency & Inference Speed Benchmark

• Physical Consistency [$T_j = 25 + TDP \cdot R_{total}$]: Random Forest achieved a mean absolute discrepancy of only 0.0080 °C (95th percentile = 0.0339 °C), while Gradient Boosting achieved 0.1467 °C (95th percentile = 0.4152 °C).
• Computational Latency Benchmark: The physics model evaluates closed-form analytical formulas in 0.0025 ms (2.5 µs). Scikit-learn tree models require 1.89 ms (Gradient Boosting) and 62.68 ms (Random Forest) per single sample due to thread dispatching and tree traversal overhead. However, in 1,000-sample batches, Gradient Boosting achieves 236,791 samples/s (comparable to physics simulator's 262,280 samples/s). Engineering takeaway: Surrogates provide transformative speedups (10^4×–10^6×) when replacing multi-hour 3D CFD meshes, but 1D analytical formulas are already computationally lightweight.

6. Correlation Analysis & Parameter Sensitivity

To establish causal engineering design rules, statistical correlations were contrasted against local physical derivatives and normalized elasticities.

6.1 Pearson vs. Spearman Correlation Matrices

Parameter Pair	Pearson (r)	Spearman (ρ)	Dominant Physical Relationship
TDP ↔ T_j	**+0.9702**	**+0.9788**	Strongest positive: Linear thermal power dissipation drives junction temperature rise.
V_{air} ↔ R_{total}	**−0.7964**	**−0.8788**	Strongest negative: Boundary layer thinning enhances convection coefficient h.
k_{tim} ↔ R_{total}	−0.2587	**−0.3566**	Moderate negative: Hyperbolic 1/k reduction in TIM conduction resistance.
TDP ↔ R_{total}	0.0000	0.0000	Zero correlation: In linear heat transfer, thermal resistance is independent of power.
Input Pairs (TDP, V, k)	0.0000	0.0000	Perfect orthogonality: Guaranteed by the full-factorial grid sampling design.

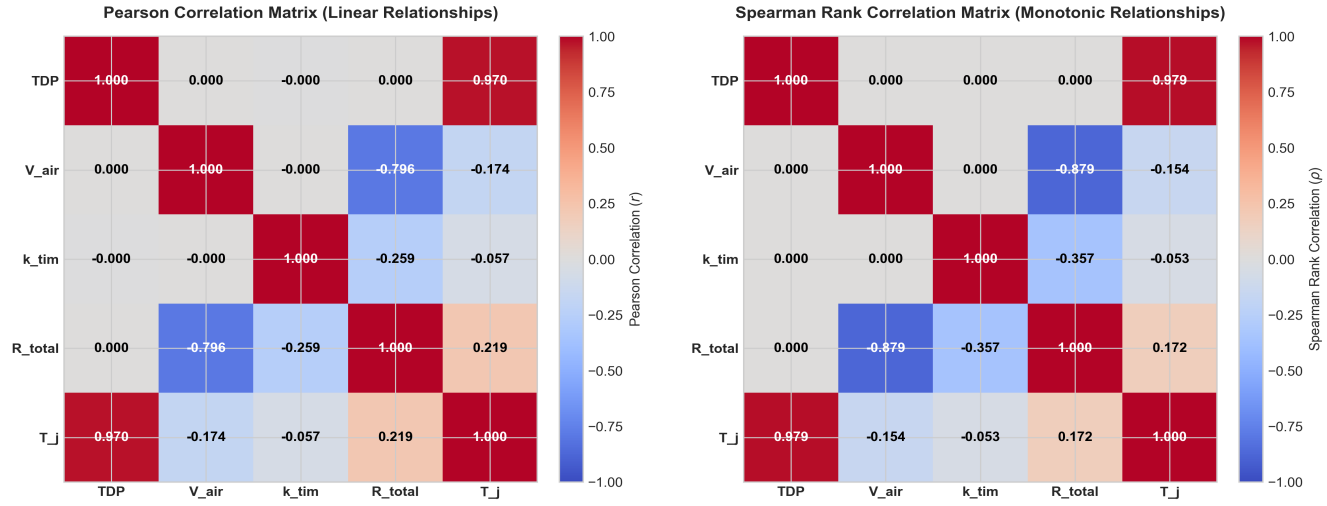


Figure 3: Pearson linear correlation (left) vs. Spearman rank monotonic correlation (right).

6.2 Local Sensitivity & Dimensionless Elasticity

Local partial derivatives and dimensionless elasticities [$\varepsilon = (x_0 / y_0) \cdot (dy / dx)$] were evaluated around the reference operating point (150 W, 1 m/s, 4 W/m·K; R_{total} = 0.3692 °C/W, T_j = 80.38 °C):

Input Parameter (x)	Nominal (x_0)	dR_{total} / dx	Elasticity ε_R	dT_j / dx	Elasticity ε_T	Primary Engineering Role
Processor TDP (Q)	150 W	0.000000 (°C/W)/W	**0.0000**	+0.369205 °C/W	**+0.6890**	**Dominates junction temperature (7× over V_air)**
Air Velocity (V_{air})	1.0 m/s	−0.052943 (°C/W)/(m/s)	**−0.1434**	−7.9414 °C/(m/s)	**−0.0988**	**Dominates thermal resistance (5× over k_tim)**
TIM Conductivity (k_{tim})	4.0 W/m·K	−0.002647 (°C/W)/(W/m·K)	**−0.0287**	−0.3971 °C/(W/m·K)	**−0.0198**	Secondary interface resistor (rapidly saturates)

7. Parametric Response Curves & ML Feature Importance

Controlled One-At-A-Time (OAT) parameter sweeps reveal the fundamental physical response modes across the operational envelope:

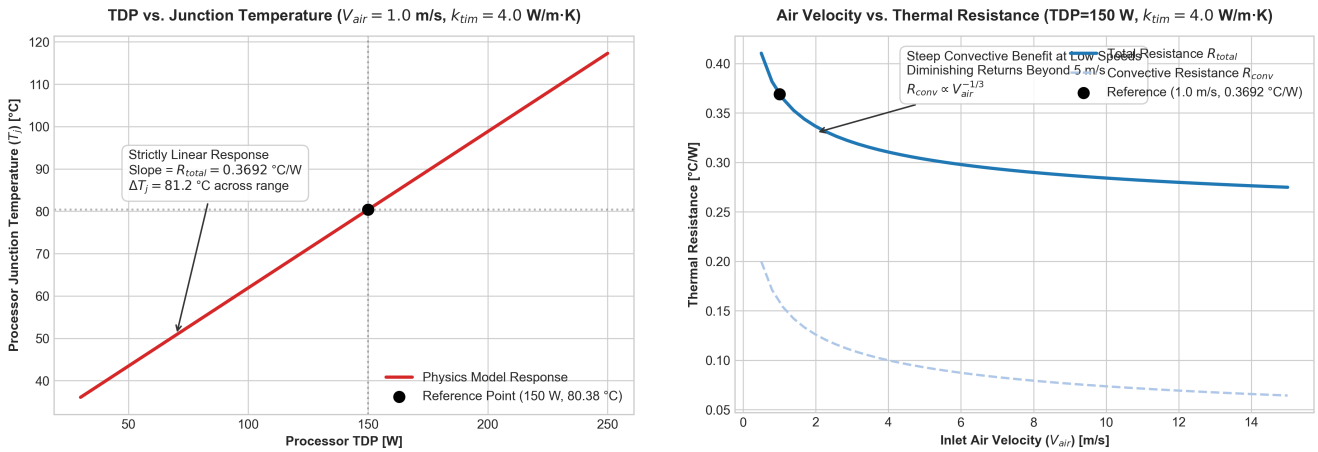


Figure 4: Linear power scaling [TDP vs. Tj] (left) vs. non-linear convective diminishing returns [V_air vs. R_total] (right).

7.1 Physics Interpretations of Non-Linear Curves

• Strict Linearity (T_j vs. TDP): At constant air velocity and TIM conductivity, thermal resistance is constant (0.3692 °C/W). Therefore, $dT_j / d(TDP) = R_{total}$ = constant across 30 to 250 W (an 81.2 °C rise from 36.1 °C to 117.3 °C with R^2 = 1.0000).

• Convective Diminishing Returns (R_{total} vs. V_{air}): In laminar channel flow, Nu is proportional to $Re^{1/3}$ and $V^{1/3}$, yielding R_{conv} proportional to $V^{-1/3}$. Increasing velocity from 0.5 to 2.0 m/s drops R_{total} by 0.0768 °C/W (saving 11.5 °C at 150 W). However, increasing from 5.0 to 15.0 m/s saves only 0.0270 °C/W (saving 4.0 °C) while increasing fan power cubically (P_{fan} proportional to V^3).

• Hyperbolic Saturation (R_{total} vs. k_{tim}): TIM resistance follows R_{tim} proportional to $1/k_{tim}$. Upgrading from 1.0 to 4.0 W/m·K saves 0.0317 °C/W (a 4.8 °C reduction). Pushing conductivity further from 4.0 to 12.0 W/m·K saves only 0.0071 °C/W because package resistance (R_{jc} = 0.20 °C/W) forms the immutable bottleneck.

7.2 Permutation Feature Importance on Held-Out Test Data

Permutation importance evaluates accuracy degradation (drop in test set R^2) when a feature is shuffled 10 times (`random_state=42`):

Model	Feature	R_{total} Drop in R^2 (mean ± std)	T_j Drop in R^2 (mean ± std)	Physics Alignment
Random Forest	TDP	−0.002295 ± 0.000181 (Zero)	**+1.920742 ± 0.036397 (Dominant)**	Correctly captures that TDP only affects temperature.
Random Forest	V_air	**+1.816740 ± 0.026540 (Dominant)**	+0.095076 ± 0.002676	Governs convective thermal resistance.
Random Forest	k_tim	+0.244195 ± 0.006888	+0.013391 ± 0.000635	Secondary interface resistance contributor.
Gradient Boosting	TDP	+0.000000 ± 0.000000 (Zero)	**+1.917903 ± 0.036463 (Dominant)**	Zero weight on resistance, dominant on Tj.
Gradient Boosting	V_air	**+1.824539 ± 0.026600 (Dominant)**	+0.094787 ± 0.002676	Highest feature importance on resistance.
Gradient Boosting	k_tim	+0.252974 ± 0.006757	+0.013644 ± 0.000663	Captures hyperbolic conduction drop.

8. Key Engineering Findings & Design Guidelines

1. Optimal Airflow Operating Window: Fan operating speeds between 1.5 m/s and 3.0 m/s yield the highest convective efficiency. Pushing air velocities beyond 5.0 m/s provides negligible thermal benefit while cubically escalating fan electrical power consumption (P_{fan} proportional to V^3) and chassis acoustic emissions.
2. TIM Specification Threshold: Standard thermal greases (3.0–4.0 W/m·K) capture 75% of available interface improvements. Specifying premium liquid metal alloys (>8.0 W/m·K) yields less than 0.005 °C/W benefit due to package junction-to-case resistance (R_{jc} = 0.20 °C/W) dominating the budget.
3. Surrogate Architecture Recommendation: Gradient Boosting is strongly recommended for production deployment over Random Forest: it achieves 5.3× lower MAE on thermal resistance, lowest RMSE on junction temperature, and requires only 149 KB of storage (a 245× reduction versus Random Forest's 36.5 MB).

9. Limitations of the Approach (Synthetic vs. Real-World Physics)

The machine learning models developed in this work demonstrate outstanding fidelity to the analytical Python simulator ($R^2 > 0.999$). However, engineering integrity requires acknowledging that high agreement with a synthetic simulator does not establish experimental accuracy against physical hardware. The underlying analytical model incorporates several simplifying idealizations:

• 1D Uniform Flow Assumption: Assumes uniform air velocity across all 60 inter-fin channels, ignoring fan hub stagnation dead zones and 3D boundary layer growth.
• Neglect of Flow Bypass: Assumes a tightly shrouded duct where 100% of the air is forced through fin channels, ignoring tip and side flow leakage common in un-shrouded server chassis.
• Omission of Thermal Radiation: Radiative dissipation can contribute 5%–15% of heat removal at elevated temperatures but is omitted from the model.
• Uniform Heat Flux Footprint: Silicon microprocessors feature non-uniform localized core power density ('hot spots') that induce 3D heat spreading resistance in the aluminum base plate, whereas the 1D model assumes uniform die heating.

Conclusion: The surrogate model accurately learns the mathematical manifold of the Python simulator, but operational deployment requires calibration against physical wind-tunnel thermocouple and infrared thermography measurements.

10. Recommendations for Future Work & Optimization

• 3D Conjugate Heat Transfer (CFD) Surrogate Training: Train multi-fidelity neural surrogates on 3D finite-volume CFD solutions (e.g., OpenFOAM, ANSYS Icepak) to capture flow bypass and local turbulence separation.
• Physics-Informed Neural Networks (PINNs): Incorporate the governing thermodynamic consistency relation [$T_j = 25 + TDP \cdot R_{total}$] directly into the neural loss function to guarantee 0.0 physical discrepancy.
• Transfer Learning on Experimental Data: Pre-train deep surrogate backbones on synthetic parameter sweeps and fine-tune on small batches of physical wind-tunnel test data.

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